Bangladesh Air Pollution

PM2.5 exposure, 1998 to 2024

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- **Executive summary**
- Exposure map
- People at risk
- Trends
- Seasonality
- Hotspots
- Drivers

Data

ACAG SatPM2.5 V6.GL.03, WorldPop 2022, HDX boundaries

Analysis

Python, DuckDB and R

Md. Ashiqur Rahman Khan

Executive summary

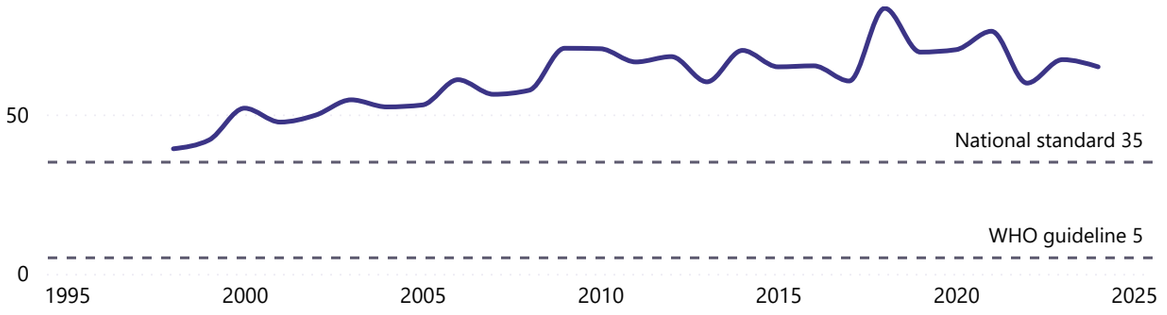
How bad is the air most people in Bangladesh breathe, and is it getting worse?

Source	Period	Unit
ACAG V6.GL.03, WorldPop 2022	1998 to 2024	µg/m³, annual, by people

PM2.5, 2020-2024	Change since 1998-2002	People above 35, 2024	Sen's slope, 1998-2024	Life expectancy lost
67.6 µg/m³	**+21.6 µg/m³**	**99.4%**	**+1.08 a year**	**6.1 years**
13.5 times the WHO guideline of 5	+47% on the 1998-2002 level of 46.1	167.3 million people in 2024	95% CI 0.65 to 1.41	Estimate per person, AQLI method

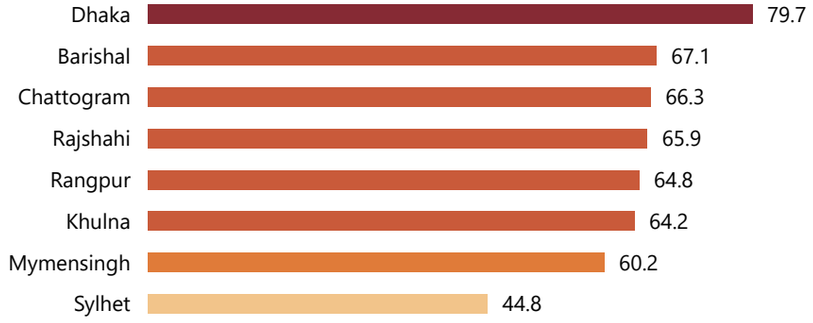
National PM2.5, 1998 to 2024

Population-weighted annual mean, µg/m³. Dashed lines: national standard 35 and WHO guideline 5. 2018 is a known spike in this data version.



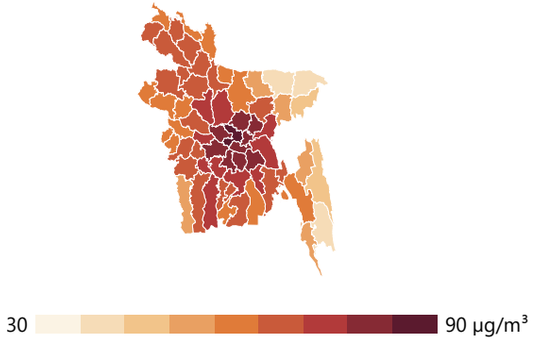
PM2.5 by division, 2020-2024

Every division is far above the national standard of 35. Colours follow the map scale.



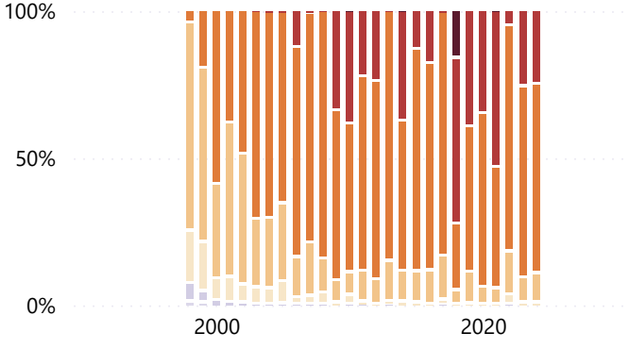
Where the air is worst, 2020-2024

By district. Right-click a district, then Drill through.



Share of people by PM2.5 band

Bangladesh, 1998 to 2024. Orange and red bands are above 35.



Key findings

From notebooks 04-05 and R scripts 06-09.

Finding	Value
Cleanest district	Sylhet 39.5
Cleanest month	July 26
Fastest-rising division	Dhaka +1.20
Upazilas with a significant rise	500 of 507
Worst district	Dhaka 83.9
Worst month	January 121
Worst upazila	Savar 85.3

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Exposure map

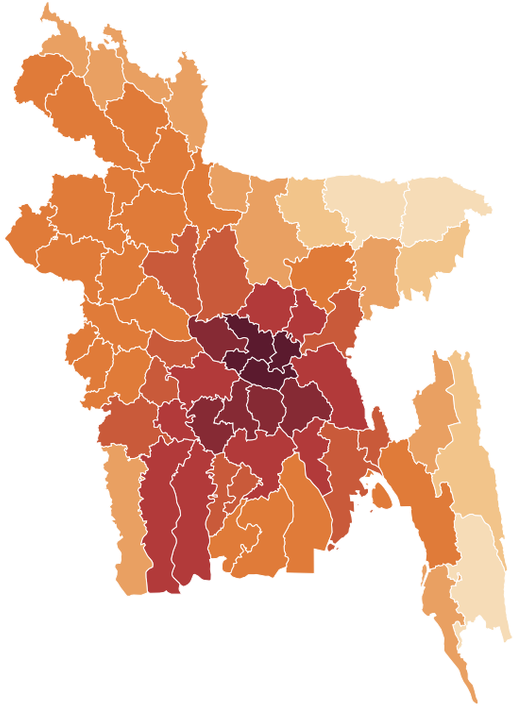
Where in Bangladesh is the air worst, and how has that changed?

Year: 2024 | Division: All | Reset

PM2.5, Bangladesh, 2024	Change, 1998 to 2024	Most polluted district	Cleanest district	Weighting gap
64.9 µg/m³	**+25.7 µg/m³**	**Dhaka 84.4**	**Sylhet 36.7**	**+5.3 µg/m³**
13.0 times the WHO guideline of 5	From 39.2 µg/m³ in 1998	Highest of 64 districts, µg/m³	Lowest of 64 districts, µg/m³	People live in the dirtier parts

PM2.5 by district, 2024

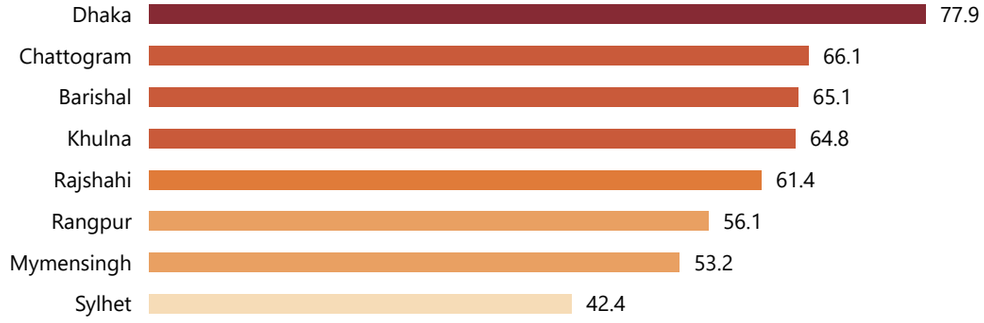
Darker: more PM2.5 where people live. The scale is fixed, so years compare. Right-click a district, then Drill through.



30 — 90 µg/m³

Division ranking, 2024

All eight divisions for the selected year, coloured on the map scale.



Most polluted districts

Top 7 in the selection. Right-click to drill.

#	District	µg/m³
1	Dhaka	84.4
2	Munshiganj	83.7
3	Narayanganj	83.4
4	Manikganj	80.5
5	Shariatpur	78.8
6	Madaripur	78.5
7	Chandpur	78.1

Cleanest districts

Bottom 7 in the selection. None meets 35.

#	District	µg/m³
1	Sylhet	36.7
2	Sunamganj	38.7
3	Bandarban	41.6
4	Netrakona	45.1
5	Moulvibazar	45.7
6	Rangamati	47.4
7	Cox's Bazar	50.4

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People at risk

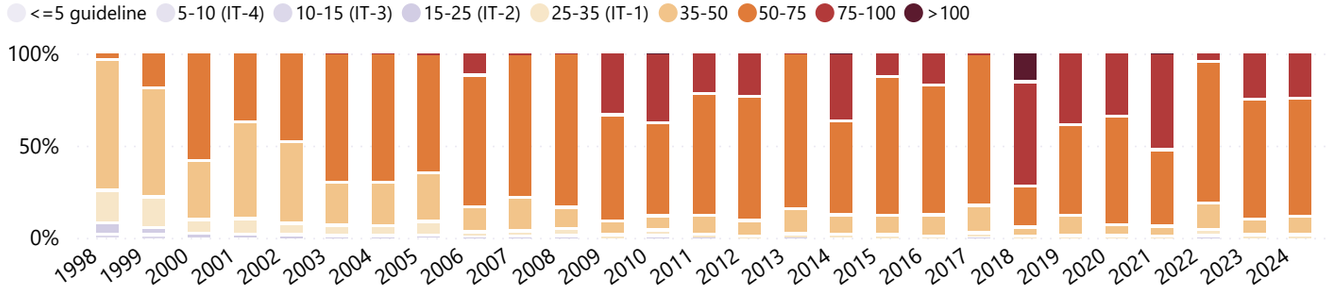
How many people live above each WHO target, and what would cleaner air be worth?

Year: 2024 | Division: All | Reset

People above 35, 2024	People above 50, 2024	People above 75, 2024	Median person, 2024	Life expectancy lost
99.4%	89%	25%	63 µg/m³	6.1 years
167.3 million people in 2024	Over 10 times the WHO guideline	42.3 million people	P10 49 to P90 82	Estimate, AQLI method, 2020-2024

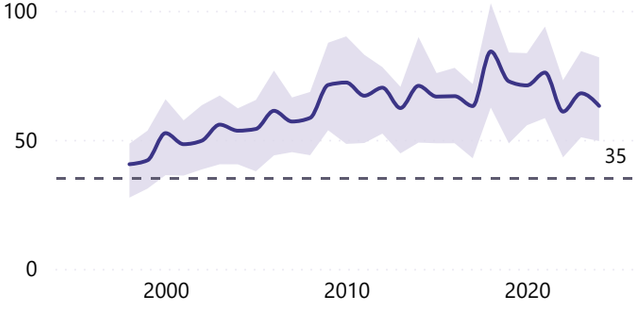
Share of people in each PM2.5 band, Bangladesh

Share of people in each WHO band, 1998 to 2024. Orange and red bands are above the national standard of 35. Bands at or below 35 hold under 1% of people.



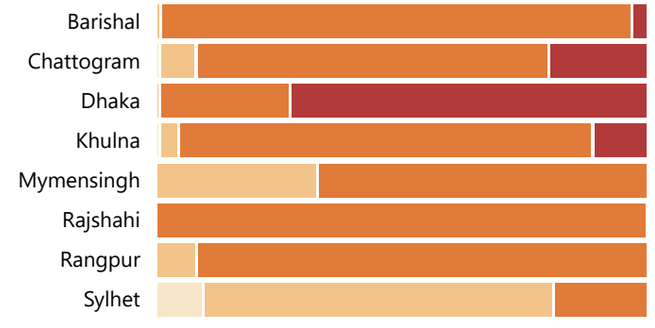
What the typical person breathes

Median person (line) and the middle 80% of people, P10 to P90 (shaded), µg/m³.



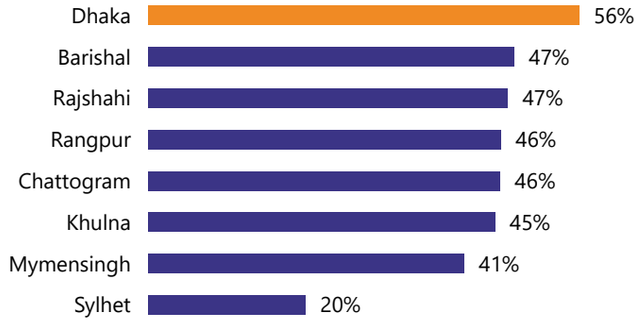
People by band and division, 2024

Each bar is 100% of the division's people.



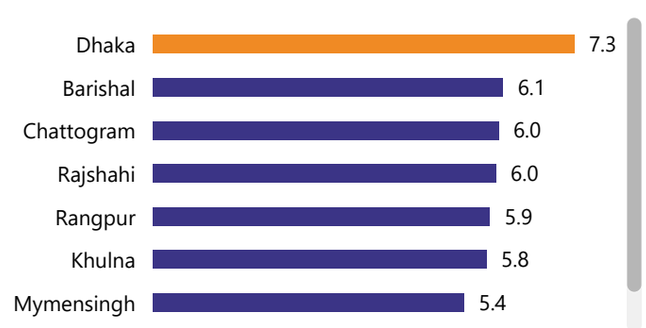
Cut needed to reach 35

Average % cut in PM2.5 needed, 2020-2024. Amber: the largest.



Life expectancy lost, estimate

Years per person vs WHO 5, AQLI method. An estimate, not measured.



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Trends

How fast did PM2.5 rise, and has it levelled off?

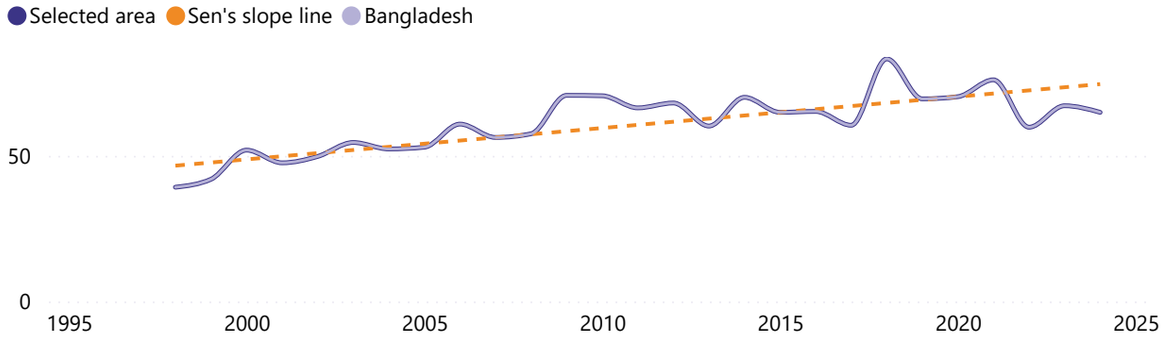
Division: All

Reset

Sen's slope, 1998-2024	Rise, 1998-2010	Rise, 2011-2024	Upazilas rising	Relative rise
+1.08 a year	+2.15 a year	+0.06 a year	500 of 507	+23% a decade
95% CI 0.65 to 1.41	p < 0.001, a significant rise	p 0.83, no clear trend	Significant after BH correction	Of the 1998-2002 level

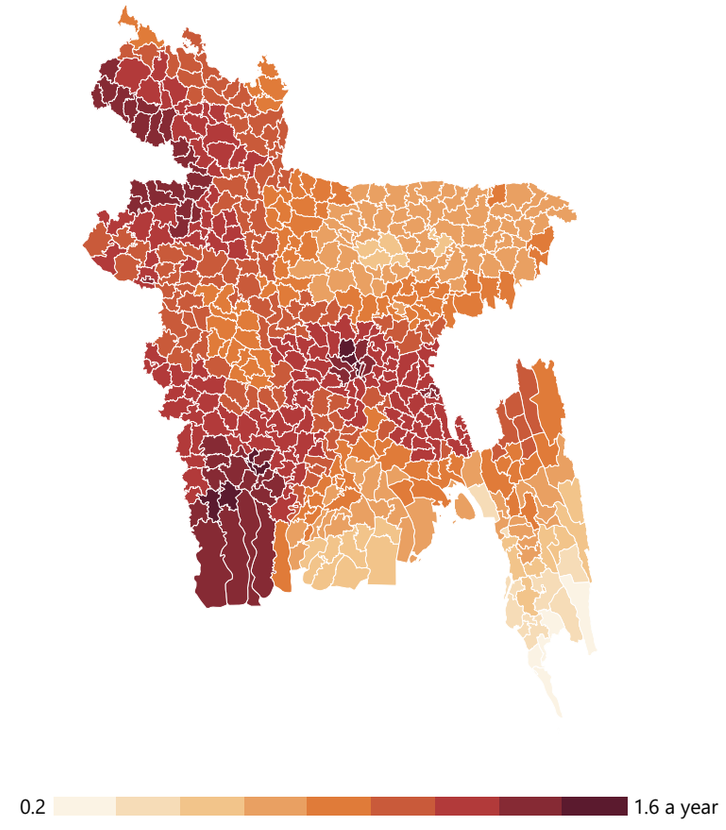
PM2.5 in Bangladesh, 1998 to 2024

Indigo: the selected area. Dashed amber: its Sen's slope line. Light: Bangladesh. 2018 is a known spike in this data version.



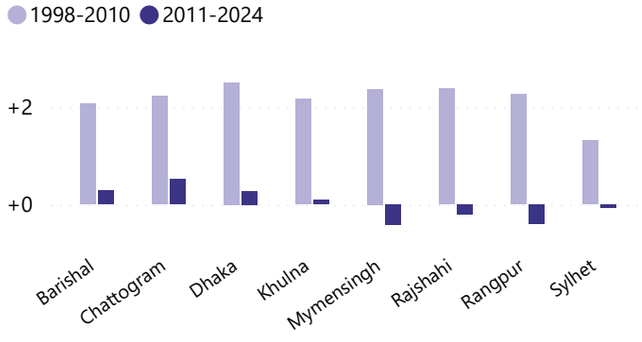
Sen's slope by upazila

µg/m³ per year, 1998-2024. Fastest rises in the north-west and on the Khulna coast.



Rise before and after 2010

Sen's slope, µg/m³ per year. Light: 1998-2010. Dark: 2011-2024.



Fastest-rising upazilas

Top 7 in the selection, µg/m³ per year.

#	Upazila	a year
1	Khulna City Corporation	+1.59
2	Dhaka South City Corporation	+1.58
3	Dighalia	+1.56
4	Rupsa	+1.54
5	Paikgachha	+1.52
6	Narayanganj City Corporation	+1.51
7	Dhaka North City Corporation	+1.51

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Seasonality

Which months are worst, and is the winter peak changing?

Division: All | Period: 2020-2024 | Reset

Worst month	Cleanest month	Seasonal swing	Largest rise	Largest relative rise
January 135	September 29	5.6x to 4.6x	December	August
µg/m³, Bangladesh, 2020-2024	µg/m³, Bangladesh, 2020-2024	Peak over trough month, 1998 vs 2024	+2.15 µg/m³ a year	+46% a decade

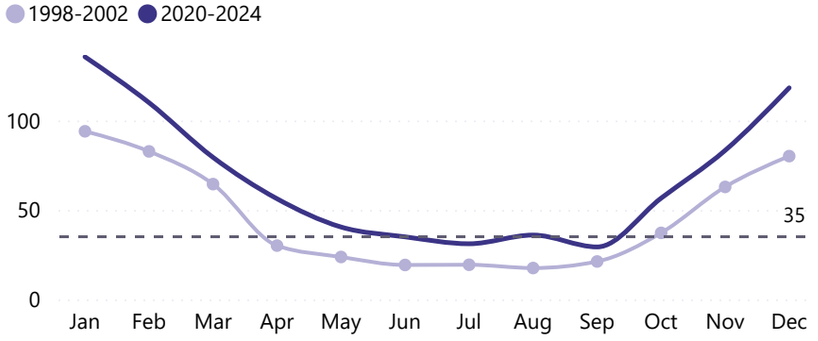
PM2.5 by month and division, 2020-2024

µg/m³. Monthly data uses the 0.1 degree grid, so it can differ slightly from the annual values.

Division	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Barishal	127	93	76	52	33	38	34	41	29	53	101	130
Chattogram	124	112	87	59	38	37	36	39	25	56	77	100
Dhaka	157	123	94	64	51	43	36	43	36	71	97	143
Khulna	124	96	72	51	39	34	29	34	28	52	95	116
Mymensingh	130	106	70	55	33	27	23	31	26	48	61	112
Rajshahi	140	113	68	50	38	28	27	30	30	52	90	125
Rangpur	145	112	72	57	37	29	26	27	28	50	78	115
Sylhet	90	81	56	40	32	25	22	25	22	38	39	68

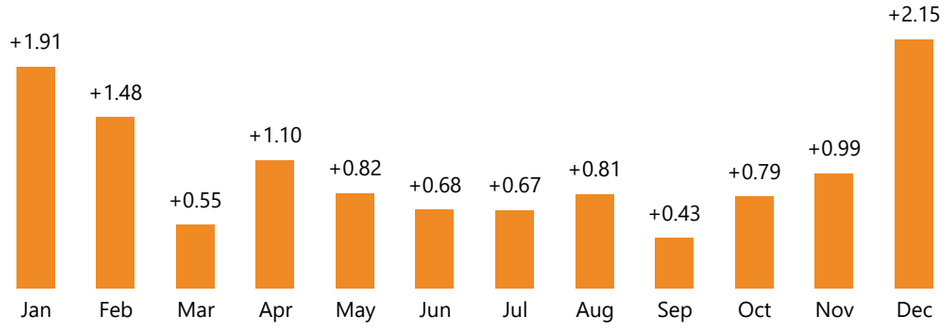
Monthly profile, then and now, Bangladesh

Monthly mean, µg/m³. Light: 1998-2002. Dark: 2020-2024.



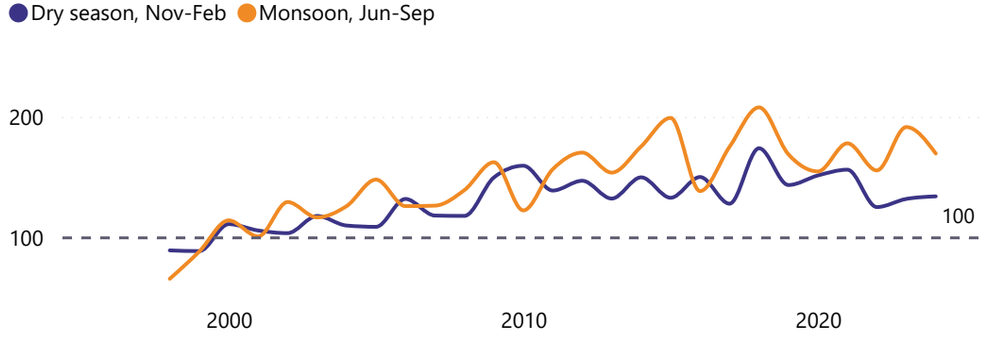
Rise by month, Sen's slope

µg/m³ per year, 1998-2024. Amber: the month that rose most.



Dry season and monsoon, indexed

1998-2002 average = 100. The monsoon rose more in relative terms.



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Hotspots

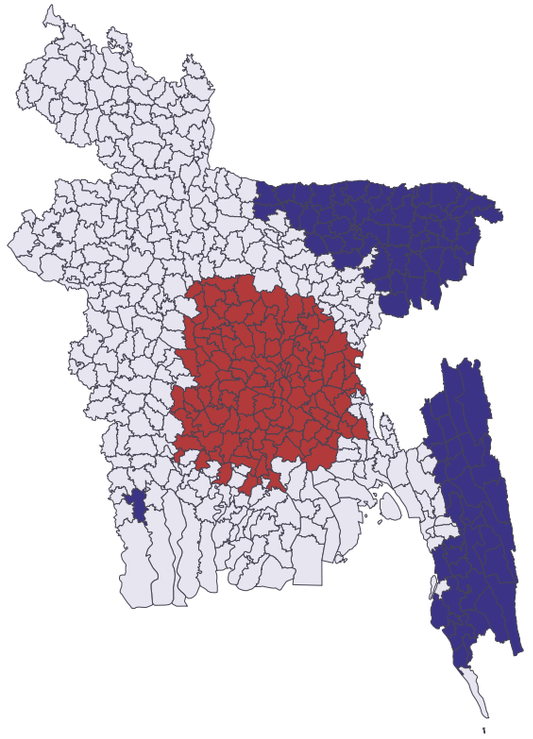
Where does pollution cluster, and how far does it spread?

Measure: PM2.5 level | Method: LISA | Reset

Moran's I, PM2.5 level	Hot cluster	Clean cluster	Gi* hot spots at 99%	Clustering fades by
0.95	**117 upazilas**	**88 upazilas**	**77 upazilas**	**About 124 km**
1 = fully clustered, 0 = random	53.9 million people, High-High	20.7 million people, Low-Low	Getis-Ord Gi*, 99% confidence	Below 0.1 at neighbour order 7

LISA clusters of PM2.5 level, 2020-2024

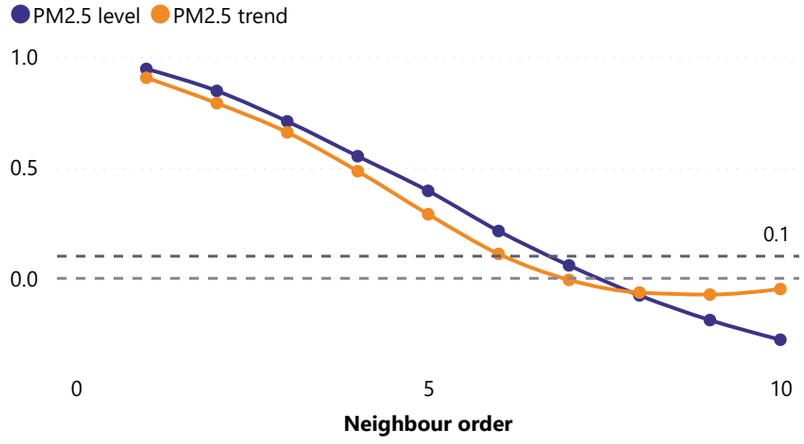
Each upazila coloured by its cluster type, p < 0.05. Use Method to switch between LISA and Gi*.



High-High or hot spot | Low-Low or cold spot | Weaker hot / cold | Not significant

How far clustering reaches

Moran's I by neighbour order. Each order adds about 18 km; order 7 is about 124 km.



Clusters and the people in them

Legend for the map. Mean value: µg/m³ for the level, µg/m³ a year for the trend.

Cluster	Upazilas	People (M)	Mean value
High-High	117	53.9	77.7
Low-Low	88	20.7	46.5
Not significant	302	93.8	65.1

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Drivers

What goes with higher PM2.5, and does machine learning agree when tested honestly?

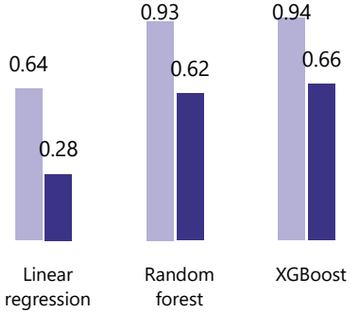
Predictor: Distance to Dhaka

Reset

XGBoost R^2, random CV	XGBoost R^2, spatial CV	Top driver, SHAP	SEM effect, Distance to Dhaka	Residual Moran's I
0.94	**0.66**	**Distance to Dhaka**	**-7.9 per SD**	**0.84 to 0.07**
Random folds let neighbours leak	Whole regions held out	Mean SHAP 3.9 µg/m³, XGBoost	OLS gave -5.4 µg/m³ per SD	OLS to spatial error model

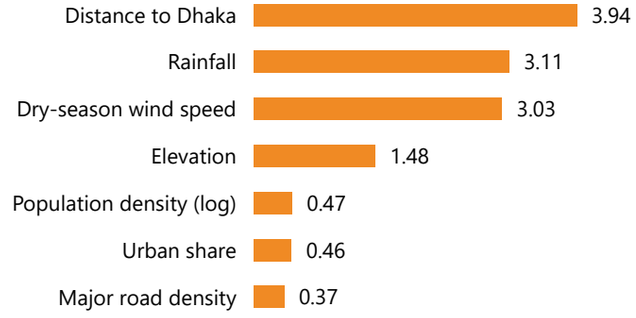
Random vs spatial CV

Out-of-fold R^2, 2020-2024 level. Light: random folds. Dark: regions held out.



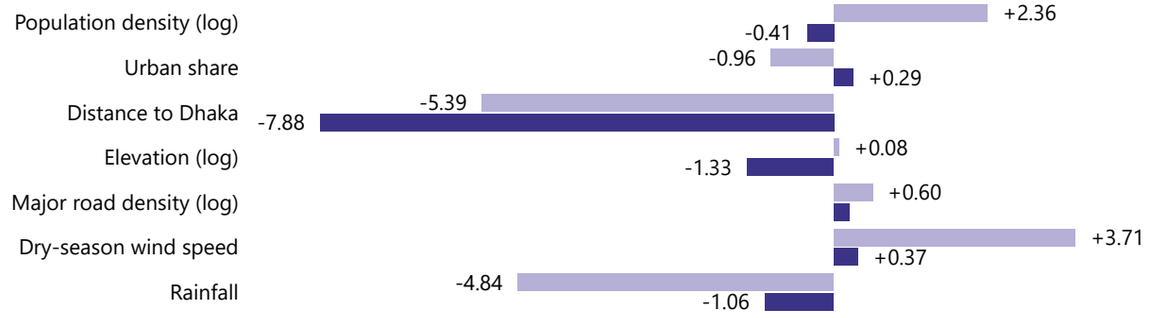
What the model leans on

Mean absolute SHAP, µg/m³, XGBoost. Amber: the largest.



Effect per 1 SD, OLS vs spatial error model

µg/m³ per 1 SD. Light: OLS. Dark: spatial error model. Associations, not causes. Hover for the 95% CI.



SHAP by upazila, Distance to Dhaka

How much this predictor pushes each upazila's predicted 2020-2024 PM2.5 up (red) or down (indigo), µg/m³. Pick another predictor at the top.

